

SAUDI ARAMCO GAS PIPELINE PROJECT - 01-SAMSS-035 QUALITY PLAN

Pipe Description: 30" (762 mm) x 14.30 mm, Grade X60 PSL 2, SAWH Pipe | Governing Standard: Saudi Aramco 01-SAMSS-035 / API Spec 5L 47th Edition

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Chemical Analysis	One analysis per heat of steel	Ladle sample	ASTM A751	C <= 0.10%, P <= 0.015%, S <= 0.005%, CE_Pcm <= 0.20	01-SAMSS-035	H/W
2	Product Chemical Analysis	Two analyses per test unit	Pipe body coupon	ISO 14284	C <= 0.10%, P <= 0.015%, S <= 0.005%, CE_Pcm <= 0.20	01-SAMSS-035	W
3	Pipe Body & Weld Tensile Test	Once per test unit (lot <= 100 pipes)	Transverse strap	ASTM A370	Rt0.5: 415-565 MPa, Rm: 520-760 MPa, Af >= 21%, Y/T <= 0.90	API 5L Tablo 7	H/W
4	CVN Charpy Impact Test at -10 °C	1 set body + 1 set weld at -10 °C	Transverse Charpy V	ASTM A370	At -10 °C: Min Average 50 Joules, Min Individual 38 Joules	01-SAMSS-035	H/W
5	Drop Weight Tear Test (DWTT)	Once per heat at 0 °C	Transverse DWTT coupon	API RP 5L3	At 0 °C: Average shear area >= 85%, No single < 60%	01-SAMSS-035	H/W
6	Guided-Bend Test	1 root + 1 face bend per test unit	Transverse weld coupon	ASTM A370	No defect > 3.2 mm in weld/HAZ after 180° bend	API 5L 9.10	W
7	Mill Hydrostatic Pressure Test	Each pipe (100% all pipes)	Full pipe length	API 5L 10.2.6	Test Pressure: 95% SMYS (149 bar), Duration: MINIMUM 15 SECONDS	01-SAMSS-035	H/W
8	Weld Seam 100% Automated UT	100% full length of helical weld seam	Full weld seam	ISO 10893-11	100% automated multi-channel UT per ISO 10893-11 Level U2	API 5L Ek E	H/W
9	Strip & Pipe Ends UT Laminar Testing	100% strip edges and 100% pipe ends	Strip edges & ends	ISO 10893-8/9	100 mm band on pipe ends; strip edges 25 mm band; no defect > 6 mm	01-SAMSS-035	W
10	Dimensional, Straightness & Magnetism	100% of all pipes	Pipe body & ends	01-SAMSS-035	Straightness <= 0.10% L; Ovality <= 0.75% D; Residual magnetism <= 2.0 mT	01-SAMSS-035	W